

EEE 466(January 2023)

Analog Integrated Circuits & Design Laboratory

Final Project Report

Group: 03

Low Noise Operational Amplifier

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Signature of Instructor: _____

Academic Honesty Statement:

IMPORTANT! Please carefully read and sign the Academic Honesty Statement, below. Type the student ID and name, and put your signature. You will not receive credit for this project experiment unless this statement is signed in the presence of your lab instructor.

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

The project involves the design and optimization of a low noise operational amplifier (op-amp) using the TSMC 18nm technology. The goal is to meet stringent specifications, including a Gain Bandwidth of 1 MHz, 10 nV/ $\sqrt{\text{Hz}}$ noise at 1 kHz, 10 pA Input Bias Current, 100 μV Voltage Input Offset, 1 mA Supply Current, 10 mA Output Current, 80 dB CMRR (at 1 kHz), and 70 dB PSRR (at 1 kHz). The design process encompasses transistor sizing, component selection, compensation, and layout design, with extensive simulation and analysis to ensure that the op-amp meets all performance criteria. The resulting op-amp will find applications in high-precision analog circuits where low noise and accurate signal processing are essential.

2 Introduction (specifications):

- **Gain Bandwidth (GBW):** Gain Bandwidth is the product of the amplifier's open-loop gain and its bandwidth. It defines the frequency range over which the op-amp can amplify signals effectively while maintaining a specified gain.
- **Noise at 1 kHz:** This represents the op-amp's noise level at a specific frequency (1 kHz). It is typically measured in nV/ $\sqrt{\text{Hz}}$ and indicates the amount of random noise added to the signal by the amplifier at that frequency.
- **Input Bias Current (I_{bias}):** Input Bias Current is the average current flowing into or out of the input terminals when no external signal is applied. It can affect the accuracy of the amplifier, especially in high-impedance applications.
- **Voltage Input Offset (V_{os}):** Voltage Input Offset is the voltage difference between the non-inverting and inverting input terminals that results in a zero output when both inputs are grounded. It can introduce an error in DC-coupled applications.
- **Current - Supply (I_{supply}):** This specification specifies the current drawn from the power supply by the op-amp. It's crucial for power management and heat dissipation considerations.
- **Current - Output (I_{output}):** Current Output is the maximum current that the op-amp can deliver to its load. It defines the op-amp's ability to drive external components or circuits.
- **Common-Mode Rejection Ratio (CMRR):** CMRR measures the ability of the op-amp to reject common-mode signals (signals that are present in-phase on both input terminals) and amplify differential signals (signals that differ between the input terminals). It is expressed in decibels (dB).

$$\frac{V_{out}}{V_{cm}} = \frac{1}{CMRR}$$

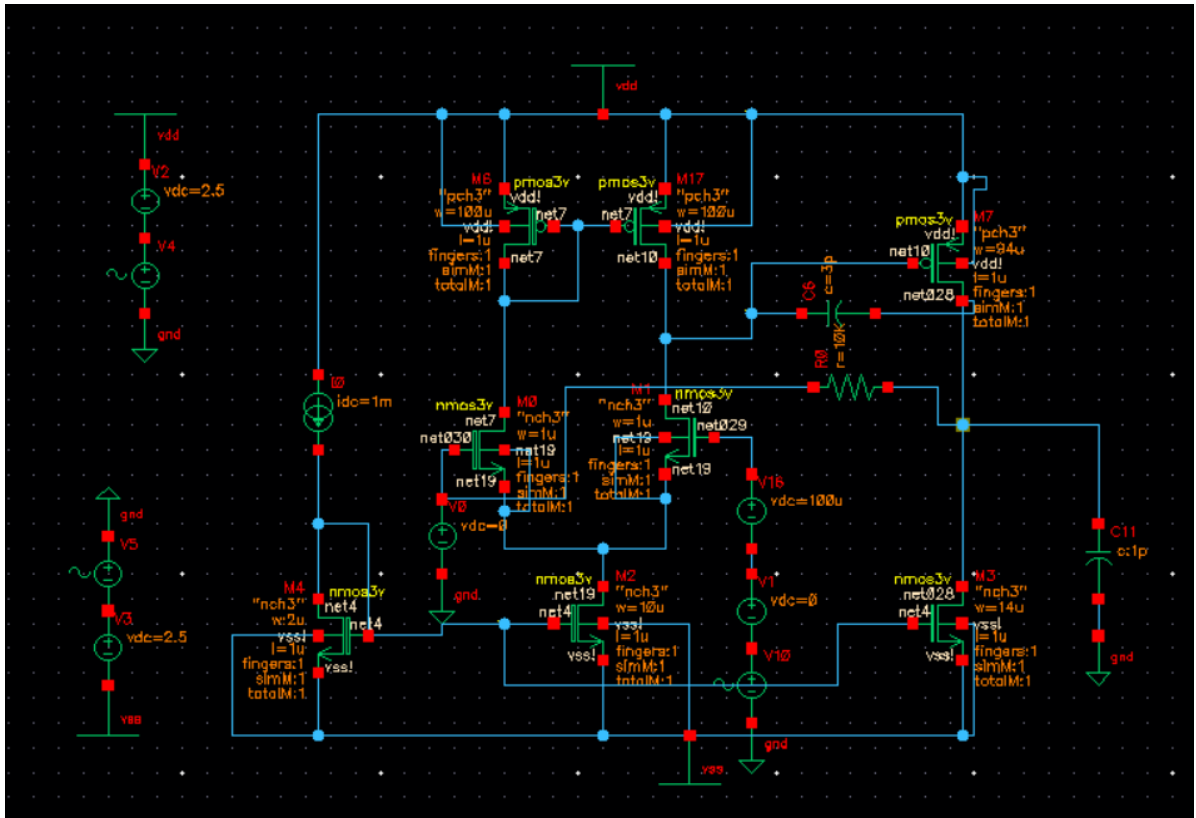
- **Power Supply Rejection Ratio (PSRR):** PSRR quantifies how immune the op-amp is to variations in the power supply voltage. It specifies how much the output will change for a given change in the supply voltage.

$$PSRR+ = \frac{\nabla V_{DD}}{\nabla V_{out}}$$

$$PSRR- = \frac{\nabla V_{SS}}{\nabla V_{out}}$$

3 Design

3.1 Simulation:



- Here, a single stage differential amplifier is created using 2 NMOS . 2 PMOS are used as diode connected loads.
- The differential amplifier is then cascaded with a single stage common source amplifier to increase the gain.
- Current mirrors are used to bias all MOSFETS precisely.
- A feedback resistance is connected between the output and negative input of amplifier to decrease noise.

4. Implementation & Results

4.1 Description:

The low noise operational amplifiers are high speed, low which enable better signal fidelity in the most demanding conditions, allowing system designers to achieve the lowest total system noise performance in applications with high source impedances.

4.2 Experiment:

As the name suggests this project basically is determined to make such design parameters so that the demanded design specifications can be met.

Simulation software cadence was used to draw the schematic and simulate the circuit. Graphs and tabulated data were obtained for each specification.

Data analysis:

supply current, supply voltage: we fixed the value of supply voltage and current as given in specification

input offset voltage: To reduce the value of input offset voltage we **reduced the W/L value** of NMOS which is used to copy current from the supply. As the formula suggests-

$$I_{out} = \left(\frac{\frac{W2}{L2}}{\frac{W1}{L1}} \right) * I_{ref}$$

Gain Bandwidth product: to increase gain we **increased the W/L** value of NMOSFETS and reduced the value of w/L of PMOSFETS. The formula-

$$A_v = \left(\frac{\frac{\mu_n W1}{L1}}{\frac{\mu_p W2}{L2}} \right)$$

Noise: to reduce noise, we added a feedback resistor between the output node and input node.

Input Current: Input current was zero at first. the feedback resistance helped to increase the input current.

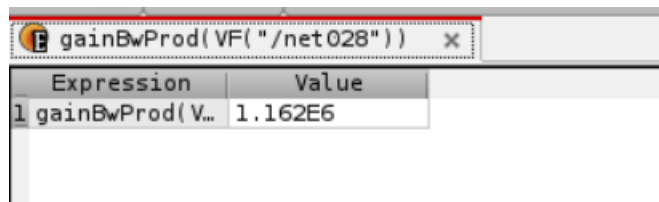
Cmrr: to improve cmrr we used current mirrors. Current mirror act as ideal current source thus improving cmrr. As the formula suggests _

$$CMRR = \frac{2 * gm^2 * R_{ss}}{gm}$$

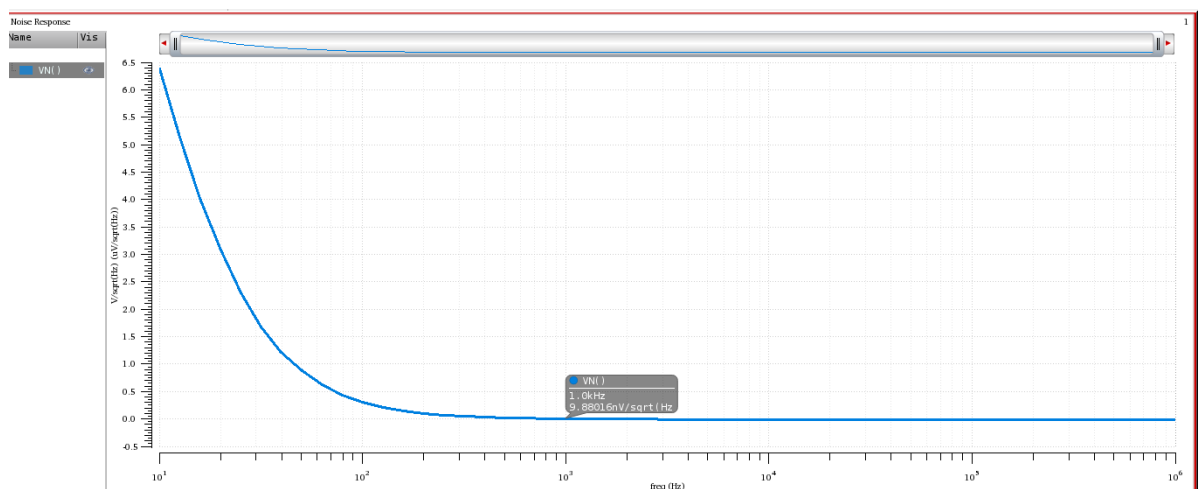
Psrr: To improve psrr we used larger capacitance at output.

4.3 Results:

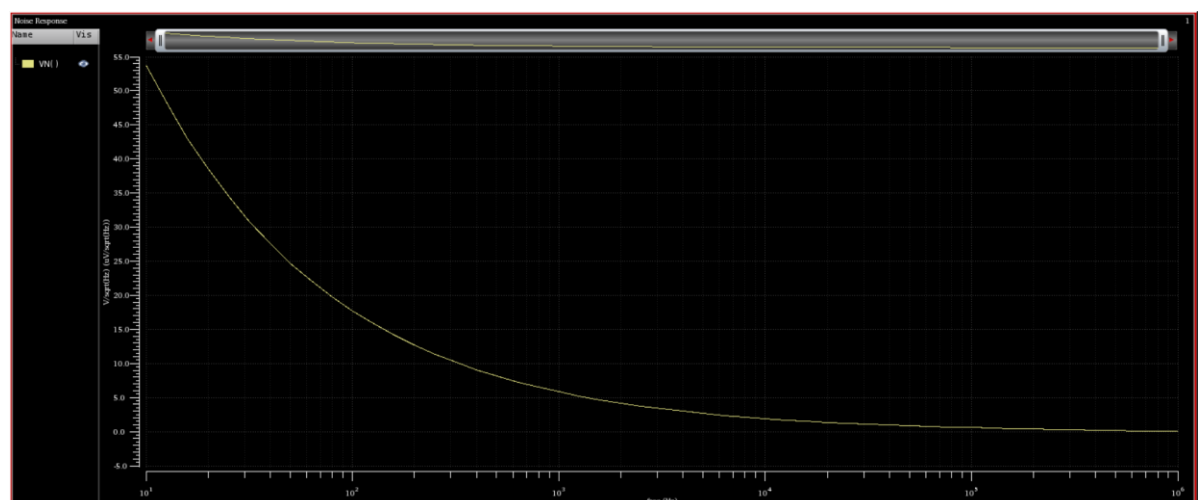
- **Specification Demanded: GainBw(1MHz)**
- **Obtained from Graph: 1.16 MHz**



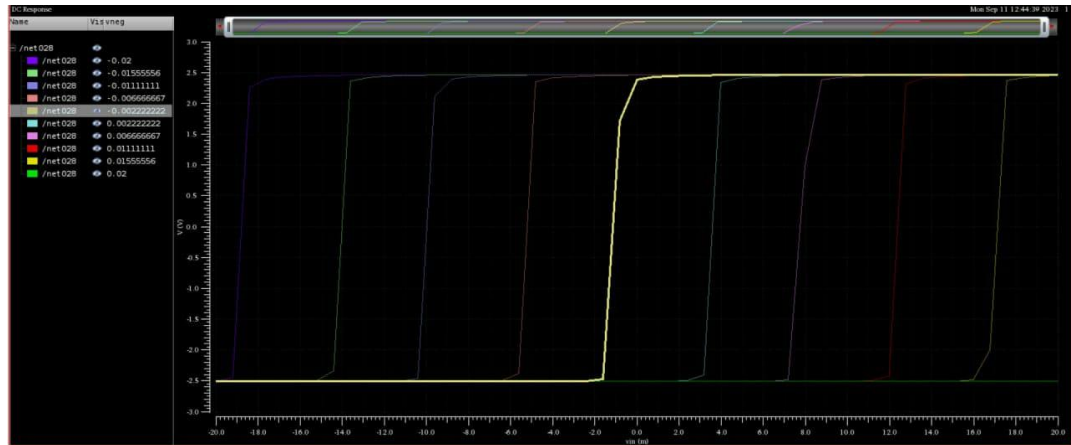
- **Demanded Specification: Noise at 1 kHz ($10 \text{ nV}/\sqrt{\text{Hz}}$)**
- **Obtained From Graph : Noise at 1 kHz ($9.88016 \text{ nV}/\sqrt{\text{Hz}}$)**



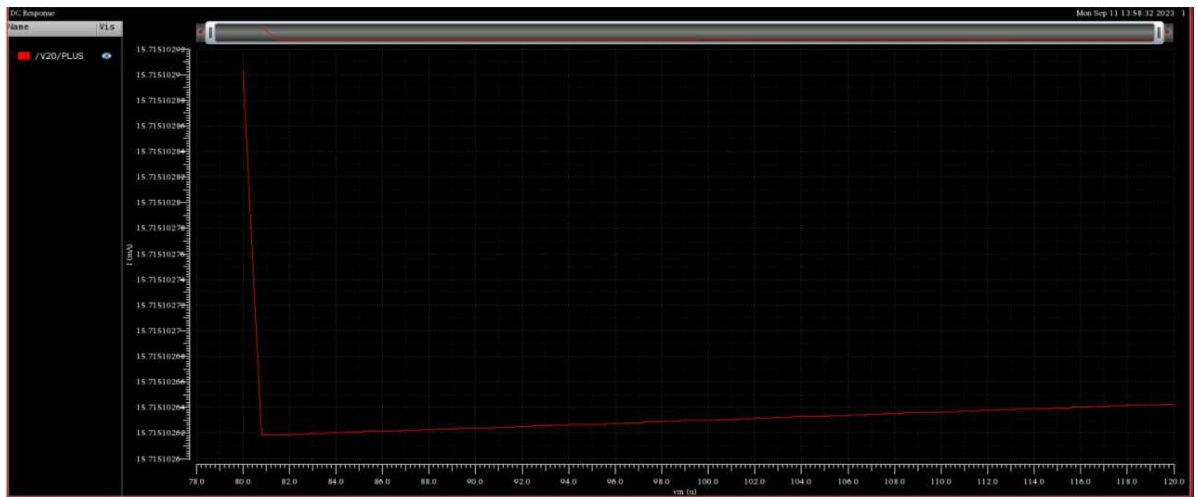
Obtained from Graph $5 \mu\text{V}/\sqrt{\text{Hz}}$ (while fixing Gain BW)



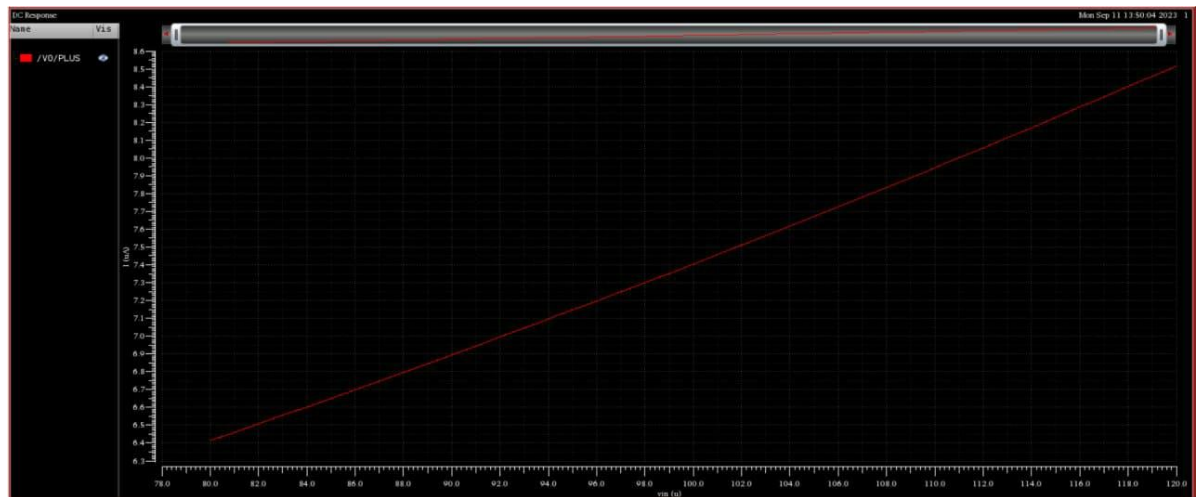
- **Demanded Specification: Voltage Input Offset ($100\ \mu V$)**
- **Obtained from Graph: ($900\ \mu V$)**



- **Demanded Specification: Current Output (10 mA)**
- **Obtained from Graph: Current Output (15 mA)**

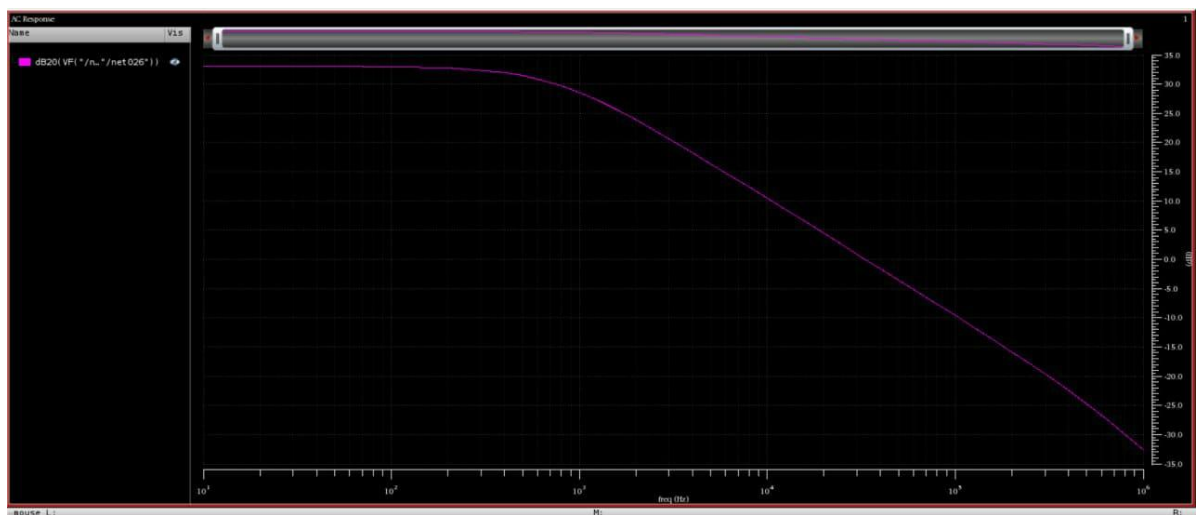


- **Demanded Specification: Current_Input Bias (10 pA)**
- **Obtained from Graph: Current_Input Bias ($.1\ \mu A$)**

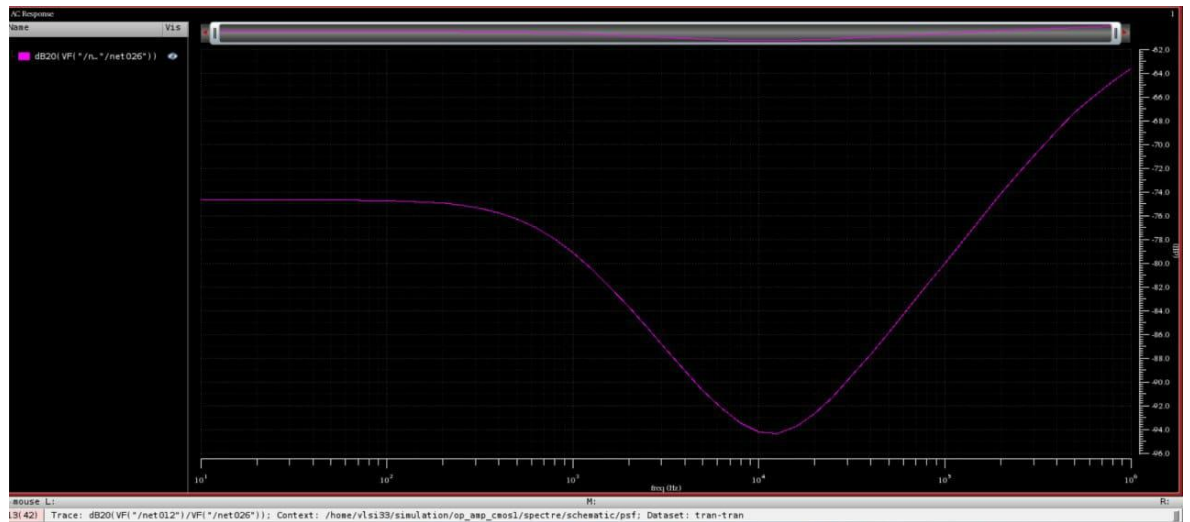


- **Demanded Specification: CMRR(at 1 kHz) -80 dB**
- **Obtained from Graph**

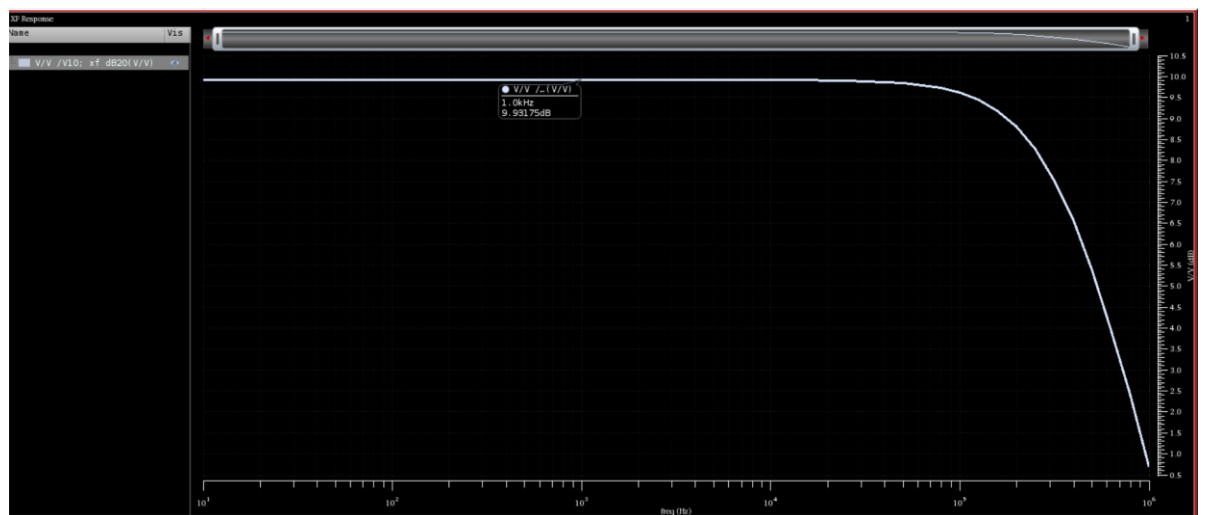
Differential Mode gain :



Common Mode Gain:



- **Demanded Specification: PSRR(at 1 kHz)- 70 dB**
- **Obtained from Graph PSRR(at 1 kHz)- 10 dB**



5. Design Analysis and Evaluation

5.1 Novelty : This project is basically determining the closest possible values so that the noise value as well as the specifications can be optimizes leading to low power consumption

5.2 Design Considerations (PO(c)):

Minimizing the width and length as much as possible . As it is known that when it will be implemented as hardware, after certain time the devices would not be biodegradable . So the lesser the size the better to handle the disposal.

5.3 Limitations of Tools (PO(e))

W and L have to .optimized but it has a certain limitation in software as well as in real life. That's why all the specifications aren't met .

6. Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member:

1806145 and 1806148 have done the simulation based works mostly.

1806143 and 1806167 also did the simulation but their main task was to get as much information for the specifications as possible .

7. Future Work (PO(l)):

- Demanded Specification optimization
- Low power Low noise Op-amp design
- Precise Specifications with better technology

8. References:

- <https://www.qorvo.com/products/amplifiers/low-noise-amplifiers>
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